GROUP 2: CLIMATE

Probabilistic Forecasting of Extreme Weather Events

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Project Context and Motivation

Title: Probabilistic Modeling of Extreme Weather Events for Risk-Informed Decision Support

- Rainfall prediction is inherently uncertain because atmospheric systems behave in a complex and chaotic manner, causing large fluctuations in rainfall intensity across different years.

- Deterministic forecasts provide only a single predicted value, which may fail to capture rare but impactful extreme rainfall events, making probabilistic analysis more useful.

https://www.bbc.com/news/articles/czx0ed2wqlvo

Project Objective and Deterministic Baseline

- Objective: To Estimate the probability that annual rainfall exceeds an extreme threshold using statistical modeling and probabilistic reasoning.

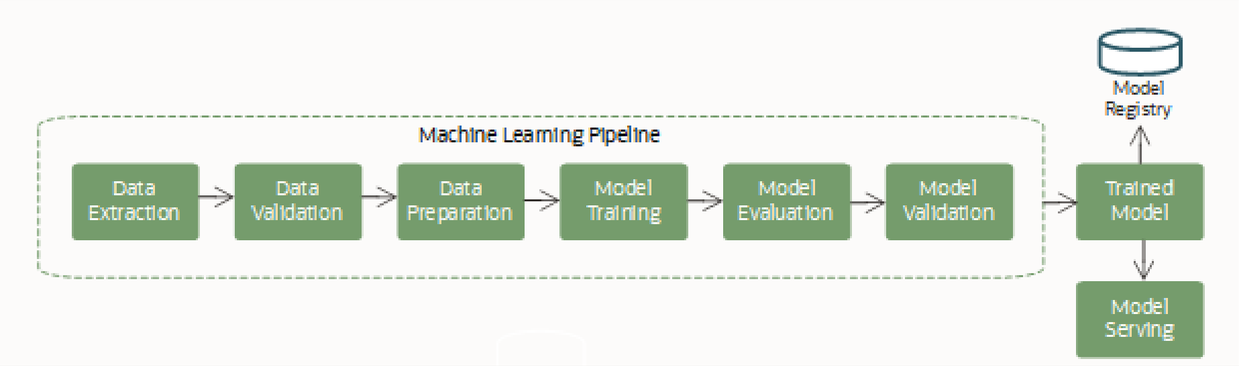
- A deterministic baseline model is implemented using logistic regression, where rainfall values are used to classify whether a year experiences an extreme rainfall event.

- This baseline approach is reasonable because rainfall intensity is a strong indicator of extreme weather events, and logistic regression provides a simple and interpretable probabilistic classifier.

System Pipeline Overview

Our probabilistic pipeline includes:

- Data Collection
- Data Preprocessing
- Uncertainty Modeling using Probability Distributions
- Extreme Threshold Definition (τ)
- Bayesian Parameter Updating
- Ensemble Probability Aggregation
- Final Risk Estimation



https://docs.oracle.com/en-us/iaas/Content/data-science/using/pipelines-about.htm

Sources of Uncertainty

- Natural variability: Rainfall patterns change significantly across years due to climate variability and complex atmospheric interactions.

- Model uncertainty: Statistical models simplify real atmospheric processes, which may introduce approximation errors.

- Data limitations: The dataset contains a limited number of observations, making rare extreme events harder to model precisely.

- In this project, uncertainty is represented through probability distributions and ensemble-based analysis.

Key Random Variables

The probabilistic model uses a few important variables to represent rainfall behavior and extreme event prediction.

- R – Annual Rainfall

Total rainfall recorded in a year from the Arunachal Pradesh dataset. It follows a lognormal distribution and is the main variable used to analyze rainfall variability.

- τ – Extreme Rainfall Threshold

Defined using the 95th percentile of the rainfall distribution. It separates normal rainfall years from extreme rainfall events.

- Y – Extreme Event Indicator

Binary variable representing extreme rainfall occurrence:
$Y = 1$ if $R > \tau$, otherwise $Y = 0$.

- θ – Model Parameters

Parameters of the logistic regression model (β_0, β_1) that determine the relationship between rainfall intensity and extreme event probability.

- $P(Y = 1)$ – Probability of Extreme Event

Final probability computed using the logistic (sigmoid) model, representing the risk of extreme rainfall.

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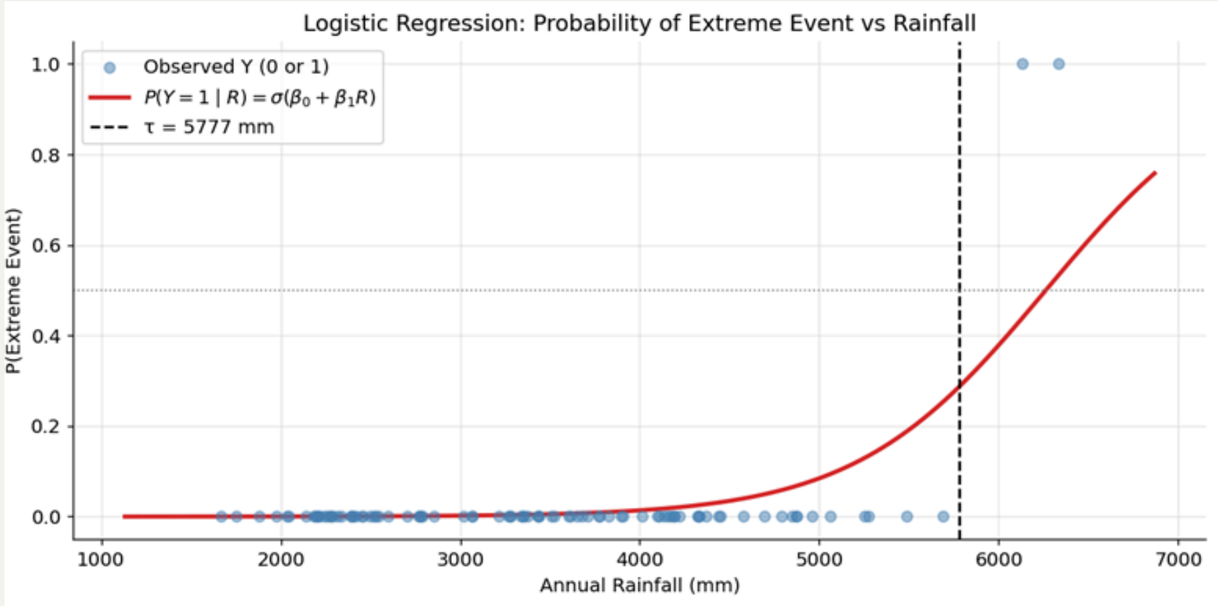
Deterministic Baseline Logic

The baseline prediction follows a fixed sequence:

- Standardize rainfall: R_{std}
- Compute probability using logistic regression: $P(Y=1|R) = \frac{1}{1+e^{-(\beta_0+\beta_1 R_{std})}}$
- Apply deterministic decision rule:

Predict extreme rainfall if probability ≥ 0.5



- Estimated Parameters:

$\beta_0 = -5.365 \quad \beta_1 = 2.025$

- Model accuracy: 98.90%

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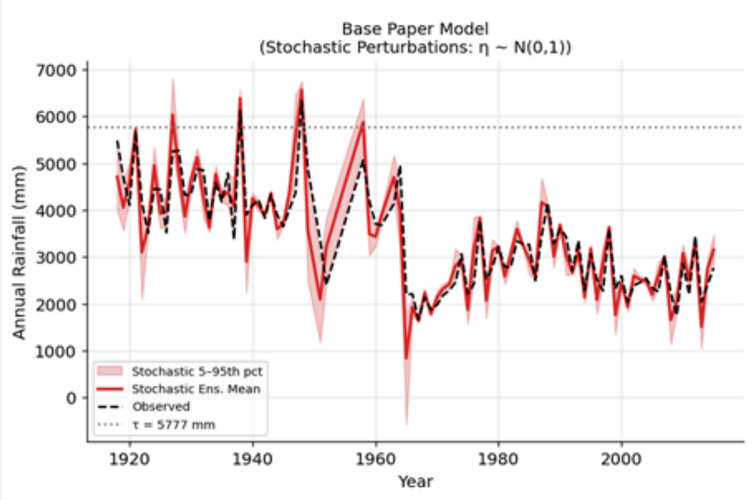
Simulation Implementation

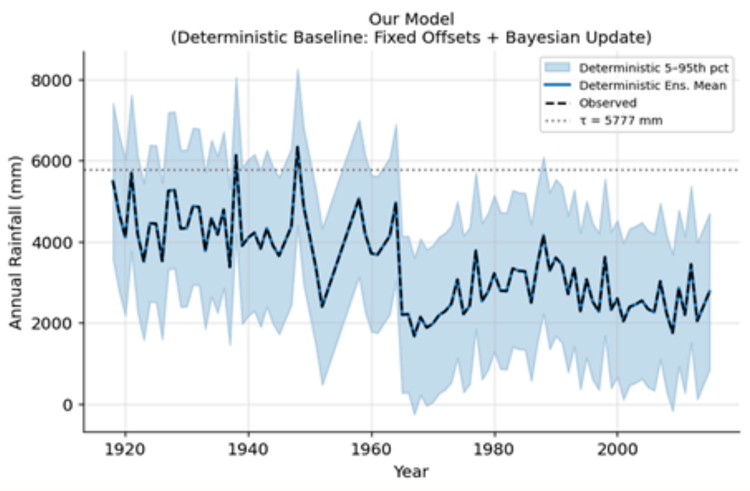
The simulation code performs the following steps:

1. Load rainfall dataset from Excel file
2. Replace "NA" with NaN
3. Remove missing values
4. Fit statistical distributions
5. Compute extreme threshold τ
6. Generate ensemble members ($M = 50$)
7. Train logistic regression using gradient descent
8. Compute ensemble probability

Results and Performance Analysis

Metric	Value
Dataset	Arunachal Pradesh Annual Rainfall, 1918–2015 (91 years)
Mean Rainfall	3418.9 mm ± 1072.9 mm
Best-fit Distribution	Lognormal (KS = 0.1004, p = 0.298)
Extreme Threshold τ	5777.2 mm (GEV, α = 0.05)
Extreme Events	2 / 91 years → $\hat{p}$ = 0.0220
Bayesian Posterior	N(3430.1, 111.37^2)
Logistic Regression	$\beta_0 = -5.365$, $\beta_1 = 2.025$
Classification Accuracy	98.90%
Ensemble P(E)	0.0853 (M = 50 equal-weight members)
Hoeffding Bound	$P(\lvert\hat{p} - p\rvert \geq 0.05) \leq 1.269$ (vacuous; n = 91 too small)





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Limitations of Deterministic Strategy

Several limitations exist in the current baseline.

- **Fixed Ensemble Structure:**
  The ensemble uses deterministic offsets rather than stochastic sampling.
- **Small Dataset:**
  Only 91 annual observations are available.
- **Synthetic Atmospheric Variables:**
  Humidity, pressure, and wind speed are simulated rather than observed.
- **Temporal Independence:**
  Each year is treated independently despite possible climate trends.

Summary and Transition to Randomized Strategies

Key Insights:

- Extreme rainfall probability is approximately 2.2%
- Logistic regression provides accurate classification.
- Ensemble probability tends to overestimate extreme events.

The model will introduce stochastic elements such as:

- random ensemble perturbations
- Bayesian parameter sampling
- improved extreme value modelling

These improvements will better capture uncertainty in extreme rainfall predictions.

Thank You